Fiscal Year:	FY 2015	**Task Last Updated:**	FY 11/08/2015
PI Name:	Delp, Michael Ph.D.		
Project Title:	Disuse Osteopenia: A Potential Vascular Coupling Mechanism		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	09/01/2014	**End Date:**	08/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per F. Hernandez/ARC (Ed., 4/6/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University School of Medicine) Delp, Judy Ph.D. (Florida State University)		
Grant/Contract No.:	NNX14AQ57G		
Performance Goal No.:			
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Task Description:

NOTE: Continuation of "Disuse Osteopenia: A Potential Vascular Coupling Mechanism," grant #NNX12AL41G. Disuse osteopenia, which is associated with habitation in a microgravity environment, is a condition of low bone density resulting in increased fragility and greater likelihood for bone fracture. Although this condition is largely attributed to the direct effects of unloading of the bone, coupling mechanisms associated with diminished bone and marrow perfusion, attenuated fluid flow within the bone, and fewer signaling molecules from vascular cells may also play a modulating role on the bone loss associated with spaceflight.

In the tail-suspended hindlimb-unloaded (HU) rat, an animal model of microgravity-induced osteopenia, hindlimb bone and marrow blood flow decrease with acute unloading (10 min) and continue to progressively decline through 28 days of unloading. The initial and chronic decrements in perfusion are primarily due to increases in osseous vascular resistance. In addition, when the HU rat hindlimb skeleton is reloaded during standing, long bone and marrow perfusion are lower and vascular resistance higher than that in standing control animals. These higher vascular resistances in HU rats could result from, 1) increased vasoconstrictor responses of bone resistance arteries, or 2) decreased endothelium-dependent and/or endothelium-independent vasodilator responsiveness of bone resistance arteries.

Specific Aims:

Aim 1. To determine whether hindlimb unloading alters the responsiveness of the principal nutrient artery (PNA) to myogenic and vasoconstrictor stimuli.

Aim 2. To evaluate the contribution of endothelium-derived NO and vasodilator prostaglandins to acetylcholine (ACh), bradykinin (BK), and flow-induced vasodilation in the femoral PNA from control and HU rats.

Aim 3. To determine whether hindlimb unloading alters the expression of endothelial NOS (eNOS), COX-1 and COX-2 mRNA and protein, as well as M2-receptor (ACh) and B2-receptor (BK) mRNA, in the femoral PNA.

Aim 4. To determine whether hindlimb unloading alters the PNA responsiveness to endothelium-independent vasodilator stimuli.

Aim 5. To determine whether hindlimb unloading, radiation exposure, and the combined effects of these treatments alters endothelium-dependent vasodilation through NO or PGI2 signaling pathways, and to determine whether this is related to changes in bone volume and indices of cancellous bone micro-architecture.

The general hypothesis of this proposal is that skeletal unloading induces alterations in skeletal resistance artery function; these alterations serve to diminish bone perfusion and ultimately disrupt mechanisms for maintaining the normal balance between bone formation and bone resorption. The specific purpose is to elucidate vasoconstrictor and vasodilator mechanisms involved in the control of bone and marrow perfusion, including endothelial signaling pathways forming nitric oxide and prostacyclin, in bone during conditions of unloading, and to determine whether deficits in bone blood flow and endothelium-dependent vascular signaling are related to unloading-induced bone loss. To accomplish these goals, the principal nutrient artery of the femur from control and 14 day HU rats will be isolated and cannulated for in vitro experimentation. Mechanisms of smooth muscle and endothelial cell function in the osseous circulation will be determined, as well as how skeletal unloading alters these vasoconstrictor and vasodilator control mechanisms (Aims 1-4). Studies will then be conducted using adenoviral gene transfer (Aim 5) to manipulate bone and marrow perfusion and endothelium-dependent vasodilation to determine whether blood flow and vascular endothelial cell signaling is coupled to disuse-associated bone remodeling in the femur (i.e., bone formation rates, mass, total, cortical and cancellous bone mineral density, and mechanical strength). These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders which occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

These studies will extend our knowledge and understanding of how skeletal unloading alters fluid and vascular coupling mechanisms, which have been suggested to modulate the balance of bone formation and degradation during microgravity. These studies may also provide new insight into the etiology of other degenerative bone disorders which occur concomitantly with cardiovascular dysfunction, such as heart disease, atherosclerosis, diabetes, and old age.

Task Progress:

Two manuscripts have been published and two more are currently under review. The first published manuscript -- Stabley JN, Prisby RD, Behnke BJ, Delp MD, "Chronic skeletal unloading of the rat femur: mechanisms and functional consequences of vascular remodeling." Bone. 2013 Dec;57(2):355-60. Epub 2013 Sep 19. http://dx.doi.org/ -- was described in the final report for the predecessor grant NNX12AL41G, and information is in that Task Book report.
Data contained in the second published manuscript (Prisby RD, Behnke BJ, Allen MR, Delp MD. Effects of skeletal unloading on the vasomotor properties of the rat femur principal nutrient artery. J Appl Physiol (1985). 2015 Apr 15;118(8):980-8 and See Bibliography section below) tested functional changes to the femoral PNA vasoconstrictor (Specific Aim #1) and vasodilator (Specific Aims #2 & #4) properties. PNA vasoconstrictor responsiveness to myogenic norepinephrine, clonidine, KCl, endothelin, and arginine vasopressin were unchanged by 14 d unloading, whereas vasoconstriction induced by phenylephrine was diminished. These data indicate that vasoconstrictor responsiveness is not enhanced in the PNA, and therefore, altered intrinsic vasoconstrictor responsiveness does not contribute to the decrease in bone and marrow blood flow and increase in vascular resistance during reloading. In addition, changes in PNA endothelium-independent vasodilation to the NO-donor sodium nitroprusside and the metabolite adenosine were unaffected by HU. However, PNA endothelium-dependent vasodilation, such as that elicited through flow and acetylcholine, were lower in HU rats relative to that in cage controls. This change in endothelium-dependent vasodilation was mediated through the NO signaling pathway and not the COX mechanism and was associated with reductions in eNOS expression but not anti-oxidant SOD-1 expression. Such changes in endothelium-dependent vasodilation could contribute to the reduction in bone and marrow blood flow and increase in vascular resistance during reloading. Furthermore, the decrements in endothelium-dependent vasodilation are significantly related to changes in cancellous bone mineral density in the femur.

Two additional studies that have been submitted for publication were supported by this grant. The first was to determine

	the effects of spaceflight on the STS-131 and STS-135 Space Shuttle missions on mandibular bone density and bone volume. The second submitted study demonstrates that hindlimb unloading, radiation exposure, and the combined effects of these two treatments diminishes endothelium dependent vasodilation, and that both endothelium-dependent vasodilation and the NO-mediated portion of endothelium-dependent vasodilation are significantly correlated with changes in bone volume and indices of cancellous bone micro-architecture. Ghosh, P, JN Stabley, BJ Behnke, MR Allen, and MD Delp. Effects of spaceflight on the murine mandible: Possible factors mediating skeletal changes in non-weight bearing bones of the head. Submitted for review. Prisby, RD, JS Alwood, BJ Behnke, JN Stabley, DJ McCullough, P Ghosh, RK Globus, and MD Delp. Effects of hindlimb unloading and ionizing radiation on skeletal muscle resistance artery vasodilation and its relation to cancellous bone in mice. Submitted for review.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Prisby RD, Behnke BJ, Allen MR, Delp MD. "Effects of skeletal unloading on the vasomotor properties of the rat femur principal nutrient artery." J Appl Physiol (1985). 2015 Apr 15;118(8):980-8. http://dx.doi.org/10.1152/japplphysiol.00576.2014 ; PubMed PMID: 25635000; PubMed Central PMCID: PMC4398884 , Apr-2015